

Omitted Variable Bias Analysis: A LaLonde Job Training Application

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1 Introduction

This report summarizes my SPA DRP project on causal inference, omitted variable bias, and sensitivity analysis, building on the DRP readings on applied causal inference, sensitivity analysis, and causal machine learning (Chernozhukov et al. 2024, 2026; Cinelli and Hazlett 2020). The project began with foundational causal inference concepts, including potential outcomes, randomized experiments, conditional ignorability, and difference-in-differences. It then moved toward omitted variable bias in regression models, sensitivity analysis using partial R^2 , and recent extensions that connect causal inference with machine learning methods.

A central lesson from the DRP is that causal inference depends not only on an estimator, but also on the assumptions that make the estimator causally meaningful. Randomized experiments use random assignment to create comparable treated and control groups. Observational studies often rely on conditional ignorability or related assumptions. Difference-in-differences designs rely on parallel trends. In all of these cases, the causal interpretation depends on assumptions about counterfactual outcomes that cannot be fully verified from observed data alone.

The main methodological focus of my project is omitted variable bias. If an unobserved variable affects both treatment assignment and the outcome, then the estimated treatment coefficient may differ from the target causal effect. Sensitivity analysis does not remove this problem. Instead, it asks how strong an omitted confounder would need to be to change the substantive or statistical conclusion.

The empirical application in this report uses the LaLonde job training data. The randomized NSW experiment provides an experimental benchmark, while CPS-based comparison groups generate non-experimental estimates that may suffer from omitted eligibility-related confounding. This setting is useful because it connects the main ideas from the DRP: causal benchmarks, comparison group construction, omitted variable bias, partial R^2 sensitivity analysis, and flexible adjustment using debiased machine learning.

2 Concepts Learned in the DRP

2.1 Potential Outcomes and Identification

The first foundational concept in the DRP was the potential outcomes framework. For each observational unit, there are two potential outcomes: $Y(1)$, the outcome if the unit receives treatment, and $Y(0)$, the outcome if the unit does not receive treatment. The individual treatment effect is

$$Y(1) - Y(0).$$

However, only one of these two potential outcomes can be observed for each unit. If $D = 1$, we observe $Y(1)$; if $D = 0$, we observe $Y(0)$. The observed outcome can be written as

$$Y = DY(1) + (1 - D)Y(0).$$

The other potential outcome is counterfactual. This is the fundamental problem of causal inference.

Because individual treatment effects are not directly observed, causal inference usually focuses on average causal effects. One common estimand is the average treatment effect,

$$ATE = E[Y(1) - Y(0)].$$

Another common estimand is the average treatment effect on the treated,

$$ATT = E[Y(1) - Y(0) \mid D = 1].$$

The distinction between these estimands matters because they refer to different target populations. The ATE concerns the overall population, while the ATT concerns the units that actually receive treatment.

The potential outcomes framework also separates estimands from identification assumptions. An estimand defines the causal quantity of interest, while identification assumptions explain when that quantity can be recovered from observed data. In randomized experiments, treatment assignment is independent of potential outcomes by design. Under consistency and overlap, comparing treated and control outcomes can identify an experimental treatment effect.

In observational studies, this comparability is not guaranteed. A common identification assumption is conditional ignorability:

$$(Y(1), Y(0)) \perp D \mid X.$$

This assumption says that, after conditioning on observed covariates X , treatment assignment is independent of potential outcomes. Another important assumption is overlap, or positivity:

$$0 < P(D = 1 \mid X) < 1 \quad \text{a.s.}$$

This means that, for almost every covariate value in the support of X , units with similar covariates have positive probability of appearing in both treatment and control groups. These assumptions are strong because they involve unobserved counterfactual outcomes. This is why causal analysis requires both statistical modeling and substantive judgment about the research design.

2.2 Omitted Variable Bias in Linear Regression

The main methodological focus of the project was omitted variable bias. Suppose the ideal or “long” regression includes treatment D , observed covariates X , and an omitted variable Z :

$$Y = \tau D + X'\beta + \gamma Z + \varepsilon.$$

If Z is unobserved, the researcher estimates a restricted or “short” regression:

$$Y = \tau_{res} D + X'\beta_{res} + \varepsilon_{res}.$$

The omitted variable bias in the treatment coefficient is

$$Bias = \tau_{res} - \tau.$$

Traditional OVB analysis shows that the bias can be understood through two residual relationships. The first is the omitted variable’s partial association with the outcome after controlling for treatment and observed

covariates. The second is the omitted variable’s partial association with treatment after controlling for observed covariates. Informally,

$$\text{OV} \approx \text{outcome relevance} \times \text{treatment imbalance}.$$

This expression is not meant as a new identifying assumption or an exact universal formula. Rather, it summarizes the logic of omitted-variable bias: an omitted variable creates bias only if it is related to both the outcome and treatment assignment, conditional on the observed covariates.

This framework clarifies why omitted variable bias is a causal problem rather than merely a statistical nuisance. In causal applications, the long regression is interpreted as the regression that would recover the target effect if the relevant confounders were observed. The short regression is what can actually be estimated. The difference between the two represents the sensitivity of the estimated treatment effect to missing confounding information.

2.3 Sensitivity Analysis with Partial R^2

Cinelli and Hazlett’s sensitivity framework extends the omitted variable bias idea into a practical set of tools for regression-based sensitivity analysis (Cinelli and Hazlett 2020). The purpose is not to prove that there is no unobserved confounding. Instead, sensitivity analysis asks what kind of omitted confounder would be required to change the research conclusion.

The traditional impact-and-imbalance parameterization is intuitive, but it can be difficult to use in practice because the parameters depend on measurement units and are hard to compare across variables. The partial R^2 framework addresses this problem by measuring confounding strength in terms of residual explanatory power.

The two main sensitivity parameters are

$$R_{Y \sim Z|D,X}^2$$

and

$$R_{D \sim Z|X}^2.$$

The first quantity measures how much residual variation in the outcome could be explained by the omitted confounder after controlling for treatment and observed covariates. The second measures how much residual variation in treatment could be explained by the omitted confounder after controlling for observed covariates.

This parameterization has several advantages. It is scale-free, can handle multiple omitted confounders, and makes it easier to compare hypothetical confounders with observed benchmark covariates. It also allows sensitivity analysis to address both point estimates and statistical significance. Robustness values summarize the minimum strength of confounding required to change a conclusion, while contour plots show how adjusted estimates or t-values change across different levels of confounding.

Benchmarking is especially important. Instead of saying abstractly that an omitted confounder is “strong” or “weak,” the researcher can compare it to observed covariates. For example, if an observed pre-treatment earnings variable explains a certain amount of residual variation in both treatment and outcome, one can ask whether an omitted confounder one, two, or three times as strong as that variable is plausible. This idea is used directly in the LaLonde application, where pre-treatment earnings variables provide interpretable benchmarks for omitted eligibility-related confounding.

2.4 Causal Machine Learning, General OVB, and DML

The DRP then moved from linear OVB to more general causal machine learning settings (Chernozhukov et al. 2026). A partially linear model with both observed and unobserved covariates can be written as

$$Y = \theta D + g(X, Z) + \varepsilon,$$

where D is the treatment, Y is the outcome, X represents observed covariates, and Z represents unobserved covariates. If Z were observed, then under consistency, regularity, and conditional ignorability assumptions, the coefficient θ could be interpreted as the target treatment effect.

In practice, Z is not observed. The researcher instead estimates a shorter model using only D and X :

$$Y = \theta_s D + g_s(X) + \varepsilon_s.$$

The omitted variable bias is then

$$\theta_s - \theta.$$

This idea is also connected to debiased machine learning (Chernozhukov et al. 2018). In the observed-covariate setting, DML estimates nuisance functions such as the outcome regression

$$g_0(X) = E[Y \mid X],$$

and the treatment regression, or propensity score,

$$m_0(X) = E[D \mid X].$$

Using estimates $\hat{g}(X)$ and $\hat{m}(X)$, DML forms residualized variables

$$\tilde{Y} = Y - \hat{g}(X), \quad \tilde{D} = D - \hat{m}(X).$$

The target treatment effect is then estimated from the residualized relationship between $\tilde{Y}$ and $\tilde{D}$. This procedure removes the parts of the outcome and treatment that are predictable from observed covariates. Orthogonalization means that the estimating equation for the treatment effect is constructed to be locally insensitive to small errors in the nuisance estimates, so that first-stage estimation errors affect the target estimate only at higher order. Cross-fitting further reduces overfitting bias by estimating the nuisance functions on one part of the data and evaluating the residuals on another part (Chernozhukov et al. 2018).

For this report, the most important takeaway is the distinction between flexible adjustment and identification. DML can reduce dependence on linear functional-form assumptions and can improve adjustment for observed covariates. However, DML still relies on the relevant identifying assumptions. It cannot remove bias from an unobserved variable that affects both treatment and outcome. Therefore, in the application, I use DML as a model-specification robustness check, not as a solution to omitted confounding.

2.5 Difference-in-Differences as a Related Extension

Another important topic in the DRP was difference-in-differences. In the basic two-period design, there are treated and control groups observed before and after treatment. Let $D = 1$ indicate the treated group and $D = 0$ indicate the control group. Let $Y_t(d)$ denote the potential outcome for an observational unit in period t under treatment status d . In the canonical two-period setup, treatment is received in the second period, so the target estimand is often the average treatment effect on the treated:

$$ATT = E[Y_2(1) - Y_2(0) \mid D = 1].$$

The key identifying assumption is parallel trends. The parallel trends for untreated potential outcomes can be written as

$$E[Y_2(0) - Y_1(0) \mid D = 1] = E[Y_2(0) - Y_1(0) \mid D = 0].$$

This assumption says that, in the absence of treatment, the treated group would have followed the same average outcome trend as the control group. Since $Y_2(0)$ is not observed for treated units after treatment, parallel trends is an identifying assumption about an unobserved counterfactual trend.

The DRP also introduced no anticipation and conditional parallel trends. In the two-period notation above, no anticipation can be written as

$$E[Y_1(1) \mid D = 1, X] = E[Y_1(0) \mid D = 1, X] \quad \text{a.s.}$$

This assumption rules out treatment effects before treatment occurs. Conditional parallel trends can be written as

$$E[Y_2(0) - Y_1(0) \mid D = 1, X] = E[Y_2(0) - Y_1(0) \mid D = 0, X] \quad \text{a.s.}$$

This condition allows treated and control groups to have different observed covariates, but assumes that untreated outcome trends are comparable after conditioning on those covariates (Chernozhukov et al. 2024; Wang et al. 2026).

A key lesson from studying difference-in-differences is that identifying assumptions can be functional-form dependent. If parallel trends holds for an outcome Y , it does not automatically hold for a transformation $g(Y)$. Therefore, even before estimation, the scale and specification of the outcome matter for causal interpretation.

The final methodological topic in the DRP was omitted variable bias in difference-in-differences designs (Wang et al. 2026). In DiD, OVB asks what happens when unobserved confounders create violations of parallel trends. The OVB-for-DiD framework studies the bias in the ATT that arises from such violations and decomposes it into interpretable components, including the strength of confounding in treatment selection, the strength of confounding in untreated outcome evolution, and the alignment between these two channels.

Although the LaLonde application below is not a DiD application, the conceptual lesson carries over. Sensitivity analysis is useful when it translates an abstract identification concern into concrete statements about the strength and plausibility of omitted confounding.

2.6 Transition to Application

The concepts above provide the methodological background for the empirical application below. The potential outcomes framework explains why an experimental benchmark is useful. The discussion of conditional ignorability and omitted variable bias explains why non-experimental CPS comparisons may fail. The partial R^2 sensitivity framework provides the main tool for evaluating how fragile the CPS-based estimates are to unobserved confounding. Finally, the discussion of DML motivates the extension that checks whether the OLS results are sensitive to linear functional-form restrictions.

In the application, I use the LaLonde job training data to connect these ideas. The randomized NSW treated/control comparison provides a benchmark estimate of the training impact, while CPS comparison groups provide non-experimental estimates that may be affected by omitted eligibility-related confounding. This setting allows me to study how comparison group construction, sensitivity analysis, and flexible adjustment affect the interpretation of non-experimental causal estimates.

3 Application: LaLonde Job Training Data

3.1 Background

The LaLonde job training data are a classic dataset in causal inference because they allow researchers to compare experimental and non-experimental estimates of the same policy intervention (LaLonde 1986). The National Supported Work (NSW) program was a job training program targeted toward disadvantaged workers. Because the NSW data include a randomized treated/control comparison, they provide an experimental benchmark for the effect of training on later earnings. LaLonde used this benchmark to evaluate whether non-experimental comparison groups, such as those drawn from the Current Population Survey (CPS), could reproduce the experimental estimate (LaLonde 1986).

This report uses the LaLonde job training data to study omitted variable sensitivity in non-experimental estimates of the NSW training effect. The randomized NSW treated/control comparison serves as the experimental benchmark because treatment assignment was randomized. In this setting, the treatment effect can be identified by comparing outcomes between treated and control units, without relying on conditional ignorability. In contrast, CPS-based comparisons are non-experimental: CPS controls were not randomly assigned and may not come from the same program-eligible population as NSW participants. Therefore, the CPS estimates rely on the assumption that, after conditioning on observed covariates, there are no remaining unobserved confounders. The main goal of this report is to examine how sensitive the CPS-based estimates are to omitted eligibility-related confounding, and whether restricting or reconstructing the CPS comparison group improves observed comparability and robustness.

3.2 Model and Identification

Following LaLonde, I refer to the randomized NSW estimate as the experimental benchmark estimate of the training impact. Since treatment assignment was randomized within the NSW experiment, the simple difference in mean 1978 earnings between treated and control units provides a causal benchmark for the NSW experimental sample. The CPS-based estimates are then evaluated by whether they can reproduce this experimental benchmark when the NSW treated group is compared with different non-experimental CPS comparison groups. I do not interpret the CPS-based estimates as average treatment effects for the broader CPS or U.S. population.

For the CPS-based non-experimental analyses, I estimate the following linear regression model:

$$re78 = \beta_0 + \tau D + X^\top \gamma + \varepsilon,$$

where $re78$ is earnings in 1978, D is the treatment indicator, and X contains observed covariates including age, education, race indicators, marital status, degree status, and pre-treatment earnings variables such as $re74$ and $re75$. The coefficient τ is the regression-adjusted estimate of the NSW training effect under the maintained identification assumptions.

The coefficient τ on D is the main quantity of interest. Since D is a binary treatment indicator, τ represents the estimated difference in 1978 earnings between NSW treated individuals and the relevant comparison group, holding observed covariates fixed. For continuous covariates, a coefficient represents the predicted change in 1978 earnings associated with a one-unit increase in that covariate. For binary covariates, a coefficient represents the predicted earnings difference when the indicator changes from 0 to 1, holding other covariates fixed.

For τ to have a causal interpretation in the non-experimental CPS comparisons, several assumptions are needed. First, consistency requires that the observed outcome equals the potential outcome corresponding to the observed treatment status:

$$Y = DY(1) + (1 - D)Y(0)$$

Second, conditional ignorability requires that, after conditioning on observed covariates, treatment status is independent of the potential outcomes:

$$(Y(1), Y(0)) \perp D \mid X$$

This is the key identifying assumption in the CPS-based analyses. It means that, conditional on X , there are no remaining unobserved factors that affect both treatment status and 1978 earnings.

Third, overlap requires that individuals with similar covariates can appear in both treatment and control groups:

$$0 < P(D = 1 \mid X) < 1 \quad \text{a.s.}$$

Although these assumptions are written in the standard potential-outcomes form, the goal here is not to estimate an ATE for the broader CPS population but to analyze whether CPS-based estimates can reproduce the NSW experimental benchmark for the NSW experimental sample.

In the CPS-based non-experimental comparisons, these assumptions are relatively strong because CPS controls were not randomly assigned and may differ from NSW participants in unobserved eligibility-related characteristics.

3.3 Constructing Experimental Benchmark

I first analyze the randomized NSW experimental data. This section serves as an experimental benchmark because the treatment and control groups come from the same experimental program and treatment assignment was randomized. I compare two specifications.

Table 1: Experimental benchmark using the original NSW sample without re74.

Mean treated	Mean control	Mean difference	N treated	N control
5976.352	5090.048	886.304	297	425

First, I compute the original experimental comparison using the no-re74 NSW sample. This gives the simple treated-control difference in mean re78, which is about 886.3. Practically, this means that individuals assigned to the NSW job training program earned about \$886 more in 1978, on average, than individuals in the randomized control group. This estimate corresponds to the original experimental comparison in LaLonde’s setting.

I next compute the experimental benchmark using the Dehejia-Wahba NSW sample that contains re74 (Dehejia and Wahba 1999). This version is especially useful for the later non-experimental analyses because the CPS comparison groups also include re74. Using the experimental sample with re74 makes the benchmark more comparable to the CPS-based specifications.

Table 2: Experimental benchmark using the Dehejia-Wahba NSW sample with re74.

Mean treated	Mean control	Mean difference	N treated	N control
6349.144	4554.801	1794.342	185	260

The estimate from the experimental data with re74 is about 1794.3, which is about twice as large as the estimate from the experimental data without re74. However, since both are based on randomized experimental

comparisons, this difference reflects the fact that the sample with re74 is a restricted subset and may represent a different experimental population. This change should be interpreted mainly as a sample-composition difference, because the Dehejia-Wahba sample excludes observations without 1974 earnings information (Dehejia and Wahba 1999). For the rest of the report, I use the experimental sample with re74 as the main benchmark for comparison with the CPS-based non-experimental estimates.

3.4 Non-Experimental CPS Analysis and Omitted Variable Bias

After establishing the experimental benchmark, I next turn to a non-experimental comparison using the NSW treated (with re74) and the CPS control group. This setup is fundamentally different from the randomized NSW comparison. The CPS controls come from an external survey sample and were not randomly assigned relative to the NSW treated group. Therefore, these controls may differ from the treated units not only in observed covariates but also in unobserved factors that affect later earnings. LaLonde’s paper emphasizes this problem by using experimental results as a benchmark to evaluate whether non-experimental econometric methods can reproduce the randomized estimate (LaLonde 1986).

A key concern is that CPS controls may not belong to the same eligible population as the NSW participants. The NSW program targeted disadvantaged workers lacking basic job skills, including groups such as ex-offenders, ex-drug addicts, etc. By contrast, the full CPS comparison group may include individuals who were not eligible. Even if two individuals have the same observed covariates, the difference in eligibility may indicate the difference in health condition, skills, motivation, etc., that are the factors influencing future earnings. In this sense, the CPS analysis relies on a strong conditional ignorability assumption.

This is exactly where omitted variable bias becomes important. If unobserved factors such as program eligibility are related to both treatment status and 1978 earnings, then the OLS coefficient on treat may combine the effect of training with pre-existing differences between NSW participants and CPS controls. Therefore, the goal of this section is to evaluate how fragile this non-experimental estimate is to omitted confounding.

In the following CPS-based analyses, I use the linear regression model introduced above.

3.4.1 Full CPS Comparison

I first compare the NSW treated group with the full CPS control group, using the sample that includes re74.

Table 3: Treatment effect estimate using the Full CPS comparison group.

Comparison group	Estimate	Std. Error	t-value	p-value	N
Full CPS	699.132	547.636	1.277	0.202	16177

The estimated coefficient is about 699, which is much smaller than the experimental benchmark and is not statistically significant.

The sensitivity analysis is implemented using the partial R^2 framework of Cinelli and Hazlett (Cinelli and Hazlett 2020) and the `sensemakr` package (Cinelli et al. 2024).

Table 4. Table for minimal sensitivity reporting: Full CPS using re75 as the benchmark covariate.

Outcome: re78						
Treatment:	Est.	S.E.	t-value	$R^2_{Y \sim D \mathbf{X}}$	$RV_{q=1}$	$RV_{q=1, \alpha=0.05}$
<i>treat</i>	699.132	547.636	1.277	0%	1%	0%
df = 16167	Bound (1x re75): $R^2_{Y \sim Z \mathbf{X}, D} = 9.3\%$, $R^2_{D \sim Z \mathbf{X}} = 0.1\%$					

Before turning to the contour plots, it is useful to explain how to read them. The horizontal axis is the partial R^2 of a hypothetical omitted variable Z with the treatment, $R^2_{D \sim Z|\mathbf{X}}$, which measures how much

additional variation in treatment assignment Z explains after conditioning on observed covariates. The vertical axis is the partial R^2 of Z with the outcome, $R^2_{Y \sim Z|D,X}$, which measures how much additional variation in the outcome Z explains after conditioning on treatment and observed covariates. In an estimate contour plot, the contour lines show the adjusted treatment-effect estimate under different strengths of omitted confounding. In a t -value contour plot, they show the corresponding adjusted t -value. The red contour line marks the confounder strengths at which the result becomes problematic, either because the adjusted estimate reaches a chosen value or because the adjusted t -value reaches the chosen significance threshold. Benchmark points compare these hypothetical confounders with observed benchmark covariates.

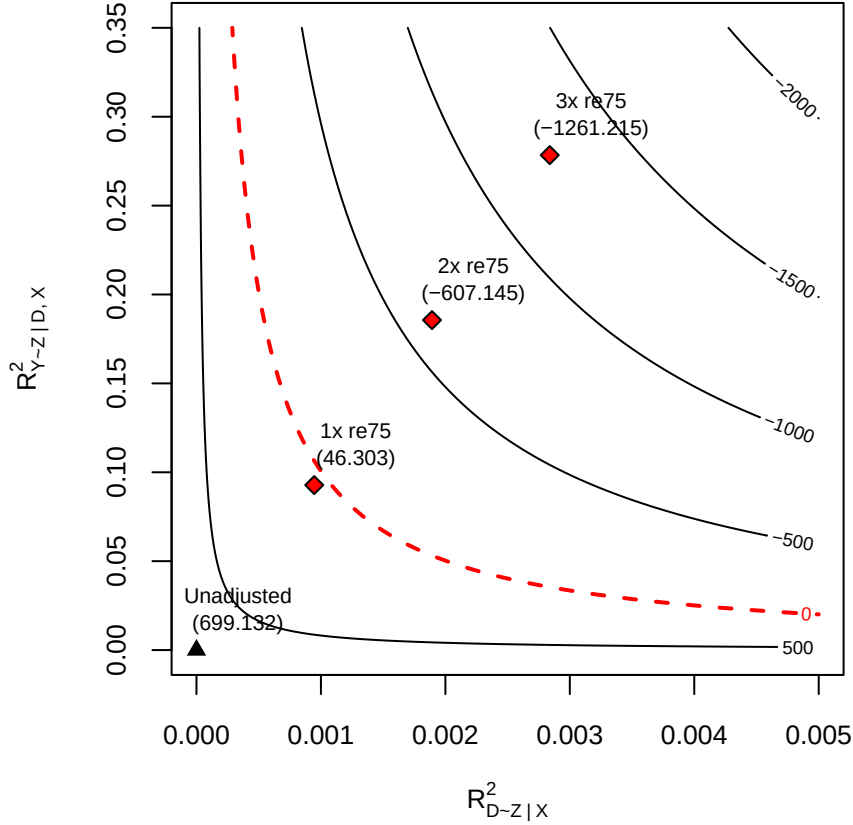


Figure 1. Full CPS estimate sensitivity contour.

The result of sensitivity analysis shows that this non-experimental estimate is very sensitive to omitted-variable bias. An omitted variable as strong as re75 can almost completely explain away the estimate.

3.4.2 Choice of Benchmark Covariate

I choose re75 as the benchmark variable because past earnings are related to an individual's labor-market attachment, which is related to future earnings. To justify the choice of re75 as the main benchmark covariate, I compare its explanatory power with other observed covariates in the full CPS model.

Table 5: Explanatory power of candidate benchmark covariates in the full CPS model.

Observed covariate	$R_{Y \sim Z X,D}^2$	$R_{D \sim Z X}^2$	$\hat{\tau} - \hat{\tau}_{-Z}$
re75	0.0848	0.0009	651.5941
re74	0.0343	0.0000	66.7542
age	0.0182	0.0001	-82.4929
education	0.0019	0.0004	-62.9449
black	0.0010	0.0782	602.0883
nodegree	0.0003	0.0023	-55.1116
hispanic	0.0001	0.0001	6.6645
married	0.0000	0.0028	14.8422

The table shows that re75 indeed has the strongest explanatory power for re78. Note that although black is more strongly related to treatment assignment than re75, re75 is much more predictive of the outcome and has a large effect on the treatment estimate when omitted. Since the sensitivity analysis is concerned with confounders that are related to both treatment and outcome, the choice of re75 as the benchmark covariate is reasonable.

3.4.3 CPS2: Restricting Toward Labor-Market Disadvantage

The full CPS comparison alone is not very informative about whether robustness improves when the comparison group is restricted toward individuals who are more similar to NSW participants. I therefore next restrict the control group to CPS2, which selects CPS individuals who were not working at the time of the 1976 survey. This restriction is motivated by the concern that many full CPS controls may not have been eligible for the NSW program. Unemployment is an observed indicator of labor-market disadvantage and may be related to weaker job readiness or basic job skills.

CPS2 should not be interpreted as a true NSW-eligible population, since it is still not drawn from the NSW applicant pool. Instead, the purpose of this comparison is to examine whether moving toward a more disadvantaged comparison group improves the robustness of the non-experimental estimate.

Note: CPS2 is very similar to, but not exactly the same as LaLonde’s subset CPS-SSA-2

Table 6: Treatment effect estimate using the CPS2 comparison group.

Comparison group	Estimate	Std. Error	t-value	p-value	N
CPS2	1172.697	645.864	1.816	0.07	2554

Table 7. Table for minimal sensitivity reporting: CPS2 using re75 as the benchmark covariate.

Outcome: re78						
Treatment:	Est.	S.E.	t-value	$R_{Y \sim D X}^2$	$RV_{q=1}$	$RV_{q=1, \alpha=0.05}$
<i>treat</i>	1172.697	645.864	1.816	0.1%	3.5%	0%
df = 2544	Bound (1x re75): $R_{Y \sim Z X,D}^2 = 5.6\%$, $R_{D \sim Z X}^2 = 0.1\%$					

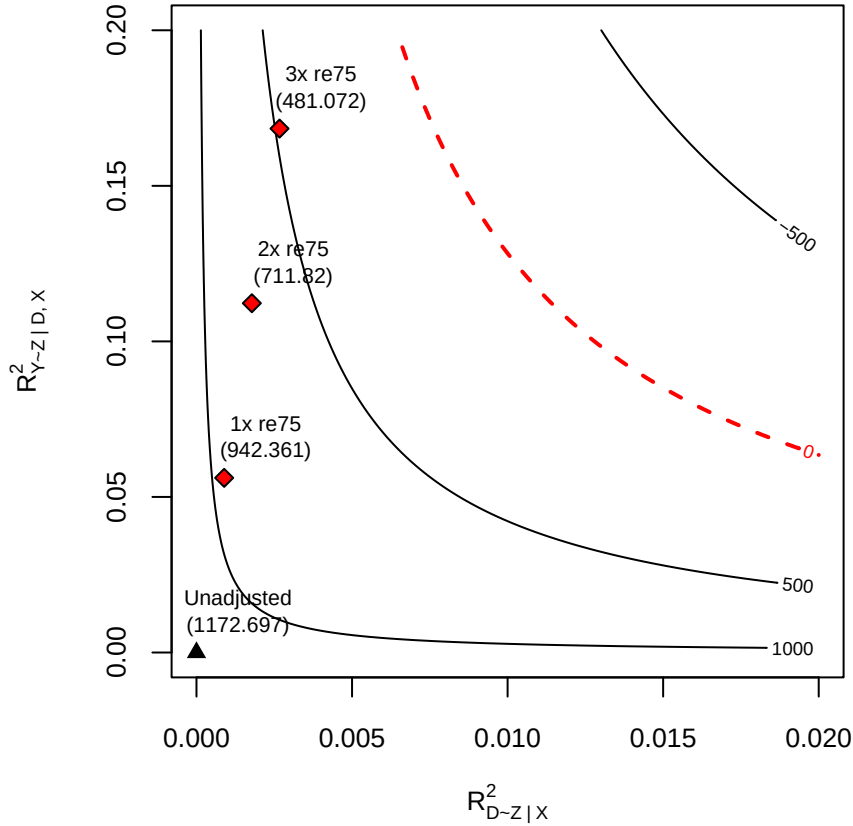


Figure 2. CPS2 estimate sensitivity contour.

The estimate is about 1172, which is larger than the estimate computed using the full CPS control group. The estimate is statistically significant at the 10% level, but not at the 5% level. The contour plot shows that a confounder three times as strong as re75 would substantially reduce the estimate, although it does not fully bring the point estimate to zero. Since the estimate is not statistically significant at the 5% level in the first place, I interpret the CPS2 result as suggestive but not robust.

This pattern is consistent with the idea that CPS2 is a more plausible comparison group. It is also consistent with the idea that CPS2 can only partially address the eligibility mismatch problem in the full CPS sample. It does not eliminate the concern that unobserved eligibility-related factors may bias the non-experimental estimate.

3.4.4 CPS3: Adding a Low-Income Restriction

I then further restrict the comparison group to CPS3. CPS3 builds on CPS2 by selecting unemployed CPS individuals whose 1975 income was below the poverty level. The motivation is that unemployment alone may still be too broad: not all unemployed individuals are disadvantaged in the same way as NSW participants. Adding a low-income restriction makes CPS3 a more economically disadvantaged comparison group and potentially closer to the NSW target population.

However, CPS3 should not be interpreted as a perfect approximation of the NSW eligible population either. Actual NSW eligibility involved more than unemployment and low income, and NSW treated participants were not necessarily all below the poverty line in 1975. Therefore, CPS3 may improve comparability along some observed dimensions, but it may also impose a selection rule that does not exactly match the NSW program.

Note: CPS3 is very similar to, but not exactly the same as LaLonde's subset CPS-SSA-3

Table 8: Treatment effect estimate using the CPS3 comparison group.

Comparison group	Estimate	Std. Error	t-value	p-value	N
CPS3	1548.244	781.279	1.982	0.048	614

Table 9. Table for minimal sensitivity reporting: CPS3 using **re75** as the benchmark covariate.

Outcome: <i>re78</i>						
Treatment:	Est.	S.E.	t-value	$R^2_{Y \sim D \mathbf{X}}$	$RV_{q=1}$	$RV_{q=1, \alpha=0.05}$
<i>treat</i>	1548.244	781.279	1.982	0.6%	7.7%	0.1%
df = 604	Bound (1x <i>re75</i>): $R^2_{Y \sim Z \mathbf{X}, D} = 0.8\%$, $R^2_{D \sim Z \mathbf{X}} = 0.1\%$					

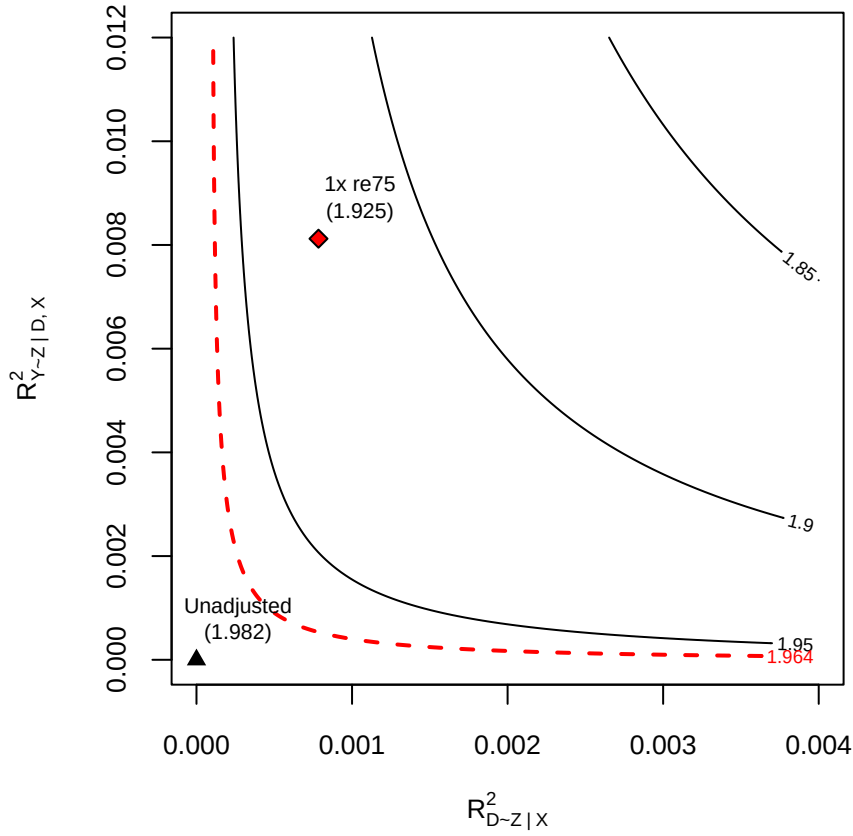


Figure 3. CPS3 t-value sensitivity contour.

The CPS3 result is the strongest among the CPS-based non-experimental estimates. CPS3 produces a larger treatment estimate of about 1548, which is statistically significant at the 5% level. However, I do not interpret this estimate as robust, because a variable as strong as **re75** would be enough to make the estimate statistically insignificant at the 5% level. This is concerning because unobserved eligibility-related covariates, such as program eligibility, may plausibly be strong.

This improvement should be interpreted cautiously. CPS3 is not a perfect approximation of the NSW eligible population, since actual NSW eligibility involved more than unemployment and low prior income, and NSW treated participants were not necessarily all below the poverty line in 1975.

3.5 Best-Case CPS Diagnostic Exercise

To further examine the role of observed comparison-group construction, I conduct a best-case CPS selection exercise. Instead of relying only on external restrictions such as unemployment or low income, I use the observed covariate distribution of the NSW randomized control group as the target and select CPS observations that most closely resemble it. This is not meant to represent a standard non-experimental design. Rather, it serves as a diagnostic benchmark: if CPS controls are selected to look like the experimental controls on observed covariates, can the non-experimental estimate reproduce the experimental benchmark?

This transition is important because it separates two ideas. CPS2 and CPS3 test whether externally motivated restrictions can improve the comparison group. The best-case CPS exercise asks whether, with knowledge of the experimental control group's observed characteristics, the CPS data contain individuals who can approximate the randomized controls. If this best-case estimate becomes more stable or more robust, it suggests that observed comparison group construction matters. Even this result would not eliminate omitted variable concerns, because similarity on observed covariates does not guarantee similarity on unobserved eligibility-related factors. However, this diagnostic has limited practical value in a genuine non-experimental evaluation, because researchers would not normally observe the randomized control group's covariate distribution. Therefore, this best-case CPS exercise should be interpreted as a diagnostic tool within this report, not as a feasible design for ordinary observational studies.

In this diagnostic exercise, I estimate a propensity score for whether an observation comes from the NSW randomized control group rather than the CPS control group. The purpose is not to estimate the probability of receiving treatment, but to measure how closely each CPS observation resembles the NSW experimental controls on observed covariates. I then rank CPS observations by this propensity score and select the top N observations, where N equals the size of the NSW experimental control group. Therefore, this procedure is better understood as propensity-score trimming or selection, rather than formal matching.

Since this selection does not guarantee covariate balance, I check balance after selection by comparing covariate distributions and binary covariate proportions across the NSW experimental controls, CPS3 controls, and selected CPS controls.

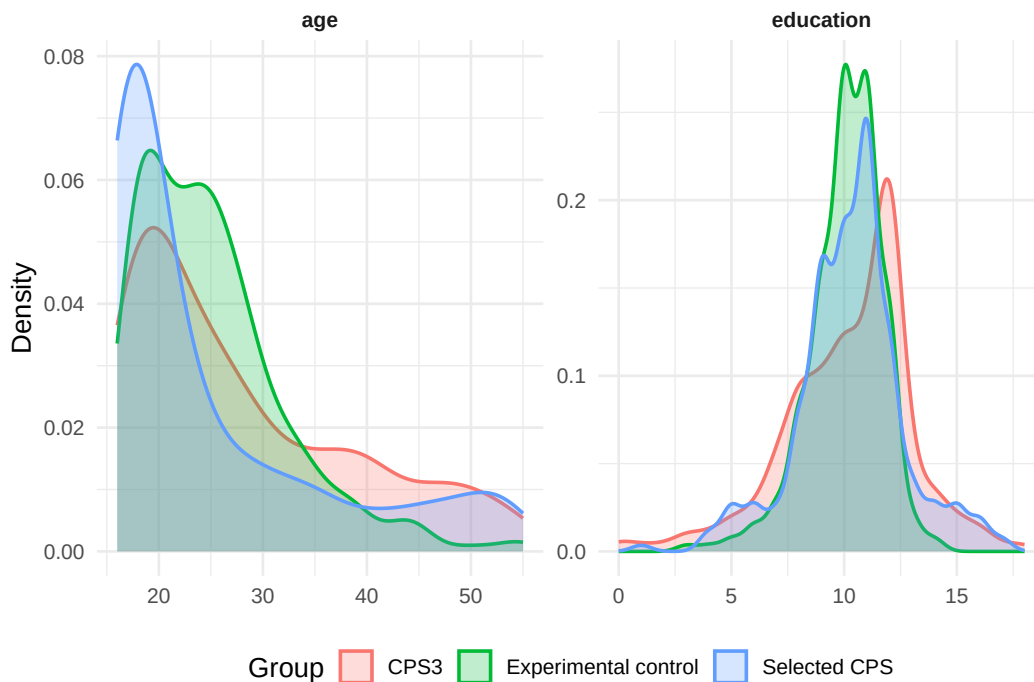


Figure 4. Continuous covariate balance across CPS3, experimental control, and selected CPS.

Table 10: Mean comparison of pre-treatment earnings variables.

group	N	Mean re74	Mean re75	% zero re74	% zero re75
CPS3	429	5619.24	2466.48	26.11	31.24
Experimental control	260	2107.03	1266.91	75.00	68.46
Selected CPS	260	1839.11	854.07	44.23	50.00

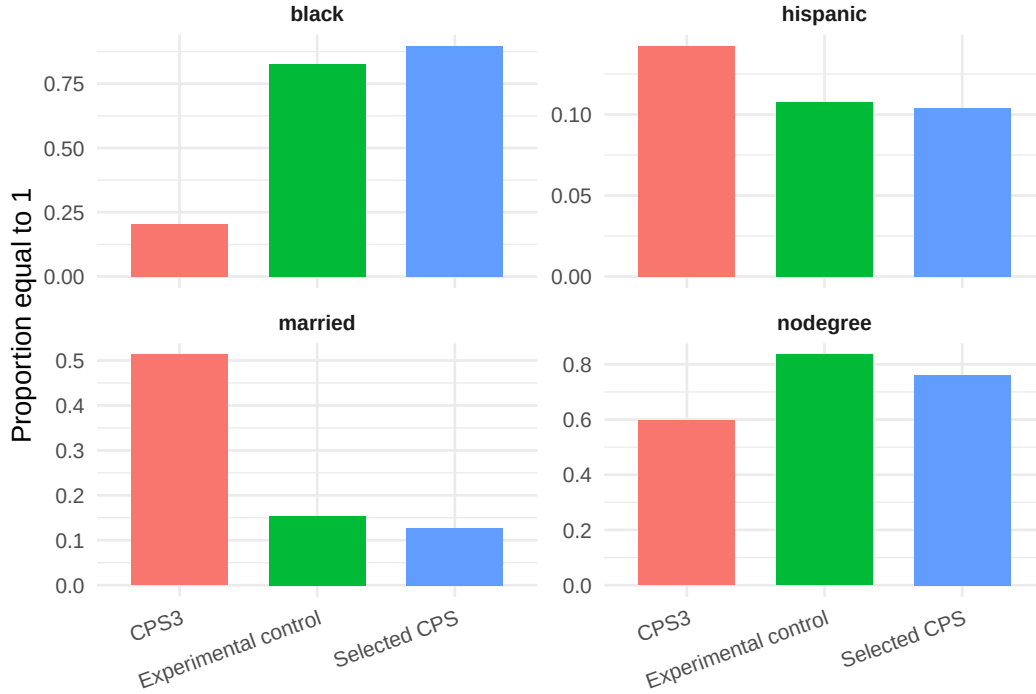


Figure 5. Binary covariate balance across CPS3, experimental control, and selected CPS.

The balance plots suggest that the selected CPS controls resemble the NSW experimental controls much more closely than CPS3. This is especially clear for the binary covariates: selected CPS is close to the experimental controls in race, marital status, and degree status, while CPS3 differs substantially on several of these dimensions. For continuous covariates, selected CPS also looks reasonably close to the experimental controls in age and education. I summarize **re74** and **re75** using means and zero-earnings shares rather than density plots because both variables contain many zero values and are highly right-skewed, making their full distributions difficult to compare visually.

The mean comparison table shows that selected CPS is much closer to the experimental controls than CPS3 in pre-treatment earnings. For example, the selected CPS group has lower mean **re74** and **re75** than CPS3 and is closer to the experimental control group. However, the table also shows that balance is still imperfect. Selected CPS has a lower share of zero earnings than the experimental controls in both **re74** and **re75**, suggesting that the selected CPS group remains somewhat more attached to the labor market before treatment. Overall, the selection procedure improves observed comparability relative to CPS3, but it does not create perfect covariate balance. Since this is a trimming procedure rather than formal matching or random assignment, remaining differences in observed and unobserved characteristics should still be interpreted as a source of potential bias.

Table 11: Treatment effect estimate using the Selected CPS comparison group.

Comparison group	Estimate	Std. Error	t-value	p-value	N
Selected CPS	1507.186	663.745	2.271	0.024	445

Table 12. Table for minimal sensitivity reporting: Selected CPS using **re75** as the benchmark covariate.

Outcome: <i>re78</i>						
Treatment:	Est.	S.E.	t-value	$R^2_{Y \sim D \mathbf{X}}$	$RV_{q=1}$	$RV_{q=1, \alpha=0.05}$
<i>treat</i>	1507.186	663.745	2.271	1.2%	10.3%	1.4%
df = 435	Bound (1x <i>re75</i>): $R^2_{Y \sim Z \mathbf{X}, D} = 0.4\%$, $R^2_{D \sim Z \mathbf{X}} = 2.3\%$					

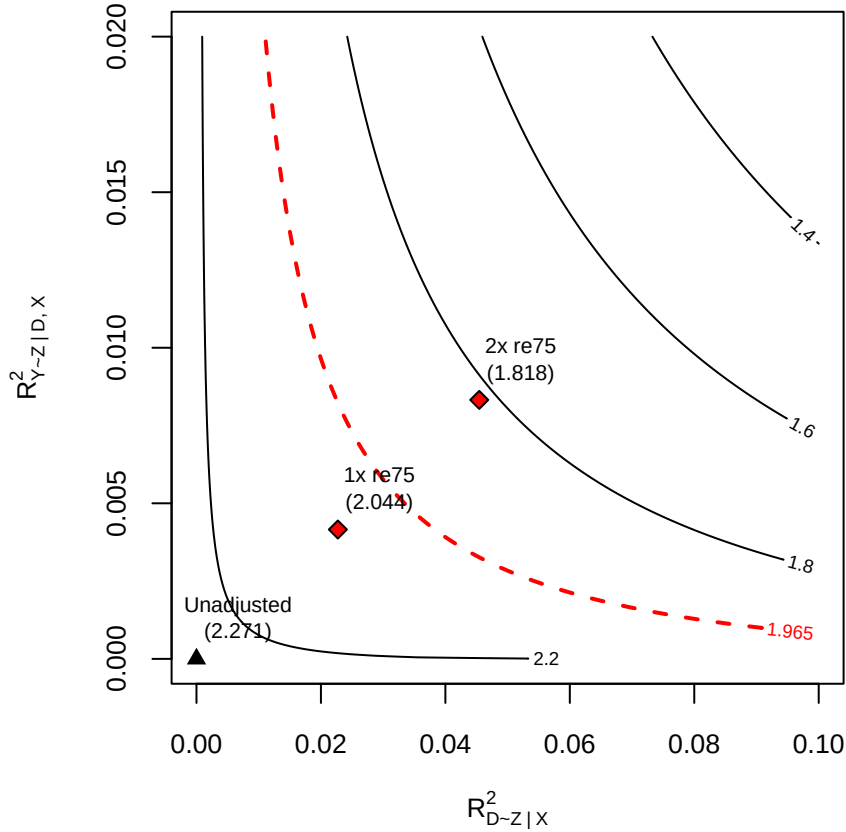


Figure 6. Selected CPS t-value sensitivity contour.

The best-case CPS exercise produces an estimate of about 1507, which is positive and statistically significant at the 5% significance level. Although this point estimate is not closer to the experimental benchmark than the CPS3 estimate, it is more robust. CPS3 gives a slightly larger estimate, but its significance is fragile because its p-value is just below 0.05 and its robustness value for maintaining significance is very small. In contrast, the best-case CPS estimate has a larger t-value and remains significant under a confounder as strong as **re75**. This suggests that selecting CPS controls to resemble the observed distribution of the NSW randomized controls does not necessarily move the estimate closer to the benchmark, but it improves the stability and robustness of the non-experimental result.

However, the estimate produced by the best-case exercise is still not the same as the benchmark estimate, which suggests that observed selection may not fully reproduce the randomized comparison. Omitted

eligibility-related confounding may still remain. This result should be interpreted as a diagnostic rather than a realistic non-experimental design. The best-case exercise uses information from the randomized NSW controls to select CPS observations, which would normally be unavailable without experimental data. Therefore, it shows that observed comparison group construction can improve robustness, but it does not prove that omitted eligibility-related confounding has been eliminated.

3.6 Extension: Debiased Machine Learning

As an extension, I use debiased machine learning (DML) to estimate the NSW training effect with more flexible adjustment for observed covariates (Chernozhukov et al. 2018). This exercise does not replace the OVB sensitivity analysis above. Instead, it checks whether the CPS-based estimates are sensitive to the linear functional-form restrictions in the OLS specifications.

In this application, I use the partially linear regression version of DoubleML. The outcome is 1978 earnings, $Y = \text{re78}$, and the treatment indicator $D = \text{treat}$ equals one for NSW treated units and zero for units in the relevant comparison group. The observed covariates X include age, education, race indicators, marital status, degree status, and pre-treatment earnings in 1974 and 1975.

The partially linear model can be written as

$$Y = \theta D + g_0(X) + \varepsilon,$$

where θ is the target treatment effect and $g_0(X) = E[Y | X]$ is the outcome nuisance function. The treatment nuisance function is

$$m_0(X) = E[D | X].$$

I estimate both nuisance functions using random forests with 500 trees: one random forest for the outcome model $g_0(X)$ and one random forest for the treatment model $m_0(X)$. I then use the partialling-out score in `DoubleMLPLR`, which residualizes both the outcome and treatment with respect to X :

$$\tilde{Y} = Y - \hat{g}(X), \quad \tilde{D} = D - \hat{m}(X).$$

The treatment effect is estimated from the residualized relationship between $\tilde{Y}$ and $\tilde{D}$. I use 5-fold cross-fitting to reduce overfitting bias, following the DoubleML partially linear regression framework (Chernozhukov et al. 2018).

This method can address misspecification in the relationship between observed covariates and earnings or treatment status. However, DML still relies on selection on observables. It can flexibly adjust for observed covariates, but it still requires the assumption that no remaining unobserved confounders affect both treatment status and the outcome. Therefore, DML does not eliminate the main concern of this report: unobserved eligibility-related confounding. I do not use DML to estimate sensitivity bounds; I use it only to compare flexible observed-covariate adjustment with the OLS estimates.

Table 13: OLS and DML treatment effect estimates across selected comparison groups.

Comparison group	OLS estimate	OLS p-value	Total N	DML estimate	DML p-value
Full CPS	699.132	0.202	16177	1195.380	0.082
CPS3	1548.244	0.048	614	939.891	0.228
Selected CPS	1507.186	0.024	445	1592.820	0.030

The DML results qualify the earlier OLS-based findings. For the full CPS comparison, the DML estimate increases from the OLS estimate of about 699 to about 1195, and the p-value decreases from 0.202 to 0.082.

This suggests that the full CPS result is somewhat sensitive to linear functional-form restrictions: because the full CPS sample is broad and heterogeneous, flexible adjustment for observed covariates can recover a larger estimated effect than the linear OLS specification. However, the DML estimate is still not statistically significant at the 5% level, and the full CPS comparison still raises serious concerns about eligibility-related mismatch.

The DML results also qualify the earlier CPS3 finding. Although CPS3 produces the strongest standard CPS-based OLS estimate, with an estimate of about 1548 and a p-value of 0.048, its DML estimate falls to about 940 and becomes statistically insignificant, with a p-value of 0.228. This suggests that the CPS3 result is sensitive to the linear specification and should not be interpreted as fully solving the comparison-group problem.

In contrast, the selected CPS estimate remains positive and statistically significant under both methods. The OLS estimate is about 1507 with a p-value of 0.024, while the DML estimate is about 1593 with a p-value of 0.030. This makes selected CPS the most stable comparison group across both OLS and DML. Therefore, the DML extension supports the broader conclusion that comparison-group construction matters, but it also shows that broad restrictions such as unemployment and low income are not sufficient by themselves. More targeted construction of the comparison group appears more stable.

However, DML still only adjusts flexibly for observed covariates, so it does not eliminate the possibility of unobserved eligibility-related confounding. The main limitation of the CPS-based estimates therefore remains the same.

3.7 Conclusion

This report examines how the robustness of estimated NSW training effects changes across experimental and non-experimental comparison groups. The experimental NSW treated/control comparison serves as the benchmark because treatment assignment was randomized. In this setting, moving from the original no-re74 sample to the smaller Dehejia-Wahba re74 sample changes the experimental estimate from about 886 to about 1794.

The CPS-based analyses show that non-experimental robustness depends strongly on how the comparison group is constructed. Full CPS produces a weak and fragile estimate, likely because many CPS controls may not resemble the NSW target population or may not have been eligible for the program. In the OLS analysis, CPS2 and CPS3 restrict the comparison group toward more disadvantaged individuals, and the estimates become stronger as the controls become more plausibly comparable. However, even CPS3 is not very robust, especially in terms of maintaining statistical significance under omitted confounding.

The best-case CPS exercise further supports this point. After selecting CPS observations that resemble the NSW randomized controls on observed covariates, the estimate becomes more statistically stable and more robust than the standard CPS-based estimates. This suggests that careful comparison group construction can substantially improve robustness. However, this exercise should be interpreted as a diagnostic rather than a fully realistic non-experimental design.

The DML extension adds one qualification to the OLS-based results. Flexible adjustment for observed covariates increases the full CPS estimate, suggesting that the broad CPS comparison may be sensitive to linear functional-form restrictions. However, the estimate remains statistically insignificant at the 5% level and still raises concerns about eligibility-related mismatch. For CPS3, the OLS estimate is statistically significant, but the DML estimate becomes smaller and insignificant, suggesting that the CPS3 result is not fully stable across specifications. In contrast, the selected CPS estimate remains positive and statistically significant under both OLS and DML. This supports the interpretation that more targeted comparison-group construction is more stable than broad external restrictions alone.

Overall, the results support the main lesson of the LaLonde data: non-experimental estimates are highly sensitive to comparison group construction. Restrictions based on unemployment, low income, pre-treatment earnings, and observed similarity to the randomized controls can improve robustness, but they cannot fully replace random assignment. The key remaining concern is eligibility-related omitted confounding, which

neither OLS adjustment nor DML adjustment can eliminate without stronger identifying assumptions or randomized assignment.

4 Overall Takeaways from the DRP

The most important takeaway from this DRP is that causal inference depends on assumptions, not only on estimators. A regression coefficient, difference-in-differences estimate, sensitivity contour, or machine learning estimate is not automatically causal. The causal interpretation depends on whether the assumptions make the observed comparison a credible substitute for the missing counterfactual outcome.

The potential outcomes framework helped me understand why this is true. Since we cannot observe both $Y(1)$ and $Y(0)$ for the same unit, causal inference always requires a way to replace the missing potential outcome. Randomized experiments do this through random assignment. Difference-in-differences does this through parallel trends. Observational regression does this through assumptions such as conditional ignorability and overlap.

A second takeaway is that omitted variable bias is not just a technical regression issue. It is a question about whether treated and control units differ in unobserved ways that matter for the outcome. The traditional OVB formula is useful because it separates bias into impact and imbalance. The partial R^2 framework makes this reasoning more interpretable by asking how much explanatory power an omitted confounder would need to have.

A third takeaway is that sensitivity analysis is diagnostic rather than corrective. It does not eliminate omitted variable bias. Instead, it helps quantify how strong an omitted confounder would need to be to change the conclusion. This is useful because it connects statistical results to substantive judgment. If only an implausibly strong confounder could overturn a result, then the result is more robust. If a relatively plausible confounder could overturn it, then the estimate should be interpreted cautiously.

A fourth takeaway is that flexible adjustment is helpful but not sufficient for causal identification. Partially linear models and DML can reduce dependence on linear functional-form assumptions and can adjust more flexibly for observed covariates. However, they still cannot solve the problem of unobserved confounding by themselves. This distinction is important: flexible prediction and causal identification are related, but they are not the same.

The LaLonde application illustrates these lessons. The randomized NSW experiment provides a credible benchmark because treatment was randomly assigned. The CPS comparisons are more difficult because the CPS controls may not come from the same eligible population as NSW participants. Restricting or reconstructing the comparison group can improve robustness, but it cannot fully replace random assignment. Overall, the application reinforces the main lesson of the DRP: causal estimates are only as credible as the comparison group and identifying assumptions behind them.

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